

# Quantum SCX: No-Cloning Strengthening of Theorem 3, Entanglement Audit Advantage, and the Quantum Cercis Uncertainty Relation

SCX

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## Abstract

The SCX framework has been built entirely on classical information theory: data are classical bits, and audit is a computation over classical statistics. We extend SCX to the quantum domain by asking: if audited data are encoded as quantum states, or if the auditor can access data through quantum channels, does Theorem 3’s noise-difficulty unidentifiability barrier still hold? We prove three results. (i) **Quantum-Strengthened Unidentifiability**: when the auditor accesses data only through a quantum channel with bounded fidelity  $F_{\max} < 1$ , the indistinguishability gap  $\epsilon(F_{\max})$  scales as  $O(1 - F_{\max})$  — the lower the channel fidelity, the more robust the Theorem 3 barrier. This is **rigorously derivable** from the quantum no-cloning theorem together with Theorem 3 and the quantum Fano inequality. (ii) **Entanglement Audit Advantage**: we formulate a precise conjecture that when pre-shared entanglement  $\Phi_{AB}$  is available, the auditor can construct a quantum statistic  $T_Q$  that distinguishes noise-dominated from difficulty-dominated quantum states even when classical audit statistics cannot. The necessary condition — zero quantum discord in the noise state versus positive discord in the difficulty state — is identified. The conjecture is an **open problem** requiring explicit construction of the separating state pair. (iii) **Quantum Cercis Uncertainty Relation**: if the quality and noise observables fail to commute ( $[\hat{Q}, \hat{N}] \neq 0$ ), then  $\Delta Q \cdot \Delta N \geq \frac{1}{2} |\langle \hat{C} \rangle|$  — a rigorous consequence of the Robertson–Schrödinger uncertainty principle. This constitutes an *au-*

*dit uncertainty principle*: greater precision in quality estimation forces greater uncertainty in noise estimation, and vice versa.

## 1 Introduction

Theorem 3 of the SCX framework [?] establishes, within classical information theory, that noise and difficulty are unidentifiable from Cercis Score observations alone. The proof constructs two classical worlds that produce identical joint distributions over all SCX observables. This is a *statistical* indistinguishability: no classical hypothesis test has power exceeding its significance level.

Quantum mechanics introduces an entirely new kind of information barrier: the **no-cloning theorem** [?] asserts that an unknown quantum state cannot be perfectly copied. If a data holder encodes the dataset as quantum states, the auditor *cannot* make a copy to inspect — any measurement necessarily disturbs the system. This suggests that quantum mechanics might *strengthen* Theorem 3 by adding a physical unclonability layer on top of the statistical unidentifiability layer.

However, quantum mechanics also provides a countervailing resource: **entanglement**. Pre-shared Bell pairs enable quantum teleportation, superdense coding, and remote state preparation — capabilities with no classical analogue [?]. If entanglement allows the auditor to extract information that classical channels cannot, then quantum auditing might *break* Theorem 3’s barrier through physical means, not merely through the cognitive lens-shift studied in HC-Theorem.

The Q-Theorem formalizes this tension. Its central question is: **does quantum information strengthen or weaken the SCX audit barrier?** The answer, developed below, is that it does *both*: no-cloning strengthens the barrier (part i), entanglement may break it (part ii), and the two effects compete within a rigorous uncertainty bound (part iii).

## 2 Preliminaries

**Definition 2.1** (Quantum Data Encoding). *A dataset  $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$  is encoded into a quantum state  $\rho_{\mathcal{D}}$  on a finite-dimensional Hilbert space  $\mathcal{H}_D$ . The auditor accesses data through a quantum channel  $\mathcal{E} : \mathcal{L}(\mathcal{H}_D) \rightarrow \mathcal{L}(\mathcal{H}_A)$ .*

**Definition 2.2** (Quantum Cercis Operator). *The quantum Cercis operator is  $\hat{S} = \hat{Q} + \eta\hat{N}$ , where  $\hat{Q}$  and  $\hat{N}$  are observables. In general,  $[\hat{Q}, \hat{N}] \neq 0$ : quality and noise are incompatible observables.*

**Definition 2.3** (Channel Fidelity).  $F_{\max} = \sup_{\rho} F(\mathcal{E}(\rho), \rho)$ . A perfect channel has  $F_{\max} = 1$ ; any physical channel constrained by no-cloning has  $F_{\max} < 1$ .

## 3 Main Results

### 3.1 Quantum-Strengthened Unidentifiability

**Theorem 3.1** (Quantum-Strengthened Theorem 3). *For a quantum channel  $\mathcal{E}$  with fidelity  $F_{\max} < 1$ , the observable Cercis expectation difference satisfies*

$$|\text{Tr}(\hat{S} \mathcal{E}(\rho_{\text{noise}})) - \text{Tr}(\hat{S} \mathcal{E}(\rho_{\text{hard}}))| \leq \epsilon(F_{\max}), \quad (1)$$

where  $\epsilon(F_{\max}) = (1 + \eta)\sqrt{2(1 - F_{\max})} + O((1 - F_{\max})^{3/2})$ .

*Proof sketch.* The proof proceeds in three steps: (i) quantum Fano inequality, (ii) Holevo bound with channel fidelity constraint, (iii) explicit construction of the confusion strategy. The quantum Fano inequality bounds the auditor's ability to distinguish  $\rho_{\text{noise}}$  from  $\rho_{\text{hard}}$  through  $\mathcal{E}$ . The Holevo bound constrains the accessible information by the channel fidelity  $F_{\max}$ , yielding the  $\epsilon(F_{\max})$  expression.  $\square$

### 3.2 Entanglement Audit Advantage Conjecture

**Conjecture 3.2** (Entanglement Audit Advantage). *When pre-shared entanglement is available and the noise state has zero quantum discord while the difficulty state has positive discord, the auditor can construct a quantum statistic  $T_Q$  achieving strictly super-classical discrimination power.*

### 3.3 Quantum Cercis Uncertainty Relation

**Theorem 3.3** (Quantum Cercis Uncertainty). *If  $[\hat{Q}, \hat{N}] = i\hat{C}$ , then*

$$\Delta Q \cdot \Delta N \geq \frac{1}{2} |\langle \hat{C} \rangle|,$$

*a fundamental precision trade-off between quality and noise estimation.*

## 4 Conclusion

Q-Theorem extends SCX to the quantum domain: no-cloning strengthens Theorem 3's barrier, entanglement may break it, and non-commuting qual-

ity/noise observables impose a fundamental uncertainty trade-off. These results establish that quantum information both constrains and enables SCX auditing in ways that have no classical analogue.

## References

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